

KINGSTON 1 W, RI, USA

WMO: 725120

Lat: 41.4782N

Lon: 71.5417W

Elev: 106

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 54797

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	9.9	-8.3	3.5	8.6	-3.3	4.6	13.3	19.3	31.9	17.1	31.3	3.9	N/A	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	86.2	73.8	83.3	72.1	80.8	70.6	76.2	83.0	74.7	80.5	73.3	78.4	6.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.1	127.8	79.8	72.9	122.6	78.0	71.6	117.0	76.5	39.6	83.2	38.1	80.6	37.0	78.3	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	13.5	11.9	-2.9	91.3	6.5	2.7	-7.6	93.2	-11.4	94.8	-15.1	96.3	-19.8	98.2
			-4.9	79.6	6.1	2.3	-9.2	81.2	-12.8	82.5	-16.2	83.8	-20.6	85.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	28.3	29.9	36.4	46.6	55.8	64.2	70.9	69.5	63.0	52.2	42.4	34.3	(6)
	DBStd	16.54	10.25	8.99	8.44	6.99	6.75	6.01	4.73	5.20	6.32	8.04	8.32	8.96	(7)
	HDD50	2645	672	564	426	147	18	0	0	0	0	70	256	492	(8)
	HDD65	6106	1137	983	885	552	295	85	8	17	112	402	678	952	(9)
	CDD50	2494	1	1	5	46	199	427	649	604	390	139	28	5	(10)
	CDD65	479	0	0	0	1	11	62	192	156	51	6	0	0	(11)
	CDH74	3282	0	0	0	25	104	477	1326	1089	239	22	0	0	(12)
	CDH80	714	0	0	0	6	19	106	321	232	29	1	0	0	(13)
(14) Wind	WSAvg	4.8	5.8	5.9	6.3	5.8	4.7	3.7	3.6	3.4	3.8	4.7	5.0	5.4	(14)
(15) Precipitation	PrecAvg	51.2	4.3	3.8	4.9	4.6	4.2	3.7	3.1	4.1	4.3	4.1	5.0	4.8	(15)
	PrecMax	70.2	12.4	8.3	12.3	13.0	9.6	14.3	7.4	12.7	10.9	14.9	13.0	11.3	(16)
	PrecMin	32.4	0.9	0.9	0.7	1.2	0.7	0.0	0.7	0.6	0.9	0.6	0.7	0.9	(17)
	PrecStd	7.9	2.3	1.7	2.4	2.4	2.2	2.9	1.9	2.7	2.1	2.4	2.5	2.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	56.1	65.7	77.1	82.7	88.0	90.3	89.3	83.7	77.5	66.2	60.4	(19)
		MCWB	54.3	50.4	54.7	63.5	70.1	74.1	76.3	75.3	74.2	66.0	60.1	57.0	(20)
	2%	DB	52.6	51.8	57.3	68.3	76.6	82.9	86.5	85.4	79.3	71.4	63.0	56.8	(21)
		MCWB	49.9	47.7	49.7	56.6	65.8	70.4	74.2	73.0	70.0	66.8	58.9	54.7	(22)
	5%	DB	48.3	48.2	53.2	62.9	72.3	79.4	83.6	82.4	76.7	68.4	60.2	53.6	(23)
		MCWB	45.5	44.3	48.1	53.9	63.0	68.3	72.6	71.5	68.8	63.7	55.8	51.4	(24)
	10%	DB	43.9	44.1	49.9	59.0	68.2	76.1	80.9	79.9	74.2	65.8	57.6	49.9	(25)
		MCWB	40.5	40.0	44.8	51.6	60.0	65.8	71.4	70.6	68.2	61.5	53.1	46.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	54.5	59.1	62.5	72.0	75.4	79.3	78.0	75.6	70.0	62.6	59.1	(27)
		MCDB	57.1	56.2	65.6	70.5	80.9	84.1	88.5	85.1	82.4	73.1	64.9	59.9	(28)
	2%	WB	50.6	50.1	52.0	58.7	68.6	72.7	76.8	75.6	73.4	67.7	60.3	55.9	(29)
		MCDB	52.1	52.3	55.4	62.0	75.0	80.4	84.7	81.0	77.7	70.6	61.9	57.1	(30)
	5%	WB	45.9	46.1	49.4	56.1	65.7	70.6	75.1	74.3	71.5	65.2	57.0	52.8	(31)
		MCDB	48.1	48.8	53.2	60.2	70.8	76.4	81.5	78.9	74.7	68.5	59.4	54.3	(32)
	10%	WB	41.3	41.7	45.2	53.2	63.1	68.6	73.7	73.0	70.0	62.9	53.0	48.8	(33)
		MCDB	43.7	45.3	48.9	56.6	67.7	73.5	79.4	77.1	73.0	65.8	56.3	51.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	17.5	17.4	18.9	18.6	18.5	17.8	18.7	19.2	19.0	18.6	16.3	(35)
		MCDBR	20.9	21.3	21.8	25.4	23.9	23.0	20.9	20.5	20.5	19.6	19.7	18.5	(36)
		MCWBR	20.4	17.1	14.4	13.2	13.4	12.6	10.2	10.6	12.2	13.6	17.5	18.1	(37)
	5% WB	MCDBR	21.5	21.4	19.0	17.4	19.8	19.5	17.8	16.6	15.1	16.3	18.5	18.2	(38)
		MCWBR	21.1	19.4	15.1	12.9	12.5	11.6	9.6	10.0	10.5	13.0	17.9	18.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.305	0.313	0.336	0.374	0.419	0.461	0.486	0.445	0.376	0.344	0.315	0.303	(40)
	taud		2.500	2.477	2.453	2.381	2.288	2.208	2.149	2.254	2.421	2.519	2.557	2.556	(41)
	Ebn at Noon		270	285	290	285	273	260	252	259	272	271	265	263	(42)
	Edh at Noon		23	28	32	37	42	45	47	41	33	26	22	21	(43)
(44) All-Sky Solar Radiation	RadAvg		598	877	1199	1548	1712	1878	1923	1726	1408	952	651	501	(44)
	RadStd		47	86	86	166	164	189	101	91	104	87	47	49	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+173	+85

Nomenclature: See separate page